

Month 2: Purple Team Batch 01

Scripting, Automation & Web Security

Week 6: Scripting & Automation in Network Security

Instructor: **Muhammad Saad**

Compiler

A **compiler** translates the **entire program at once** into machine code before running it.

How it works:

Source Code → Compiler → Machine Code → Program Runs

Characteristics:

- Translates the **whole program at once**
- Produces an **executable file**
- Runs **faster after compilation**
- Shows **all errors after compiling**

Examples of compiled languages

- C
- C++
- Go

Example:

When you write a C program, it must be compiled first before running.

Interpreter

An **interpreter** translates and executes the program **line by line**.

How it works:

Source Code → Interpreter → Runs line by line

Characteristics

- Executes **one line at a time**
- **No separate executable file**
- Stops immediately if an **error occurs**
- Usually **slower than compiled programs**

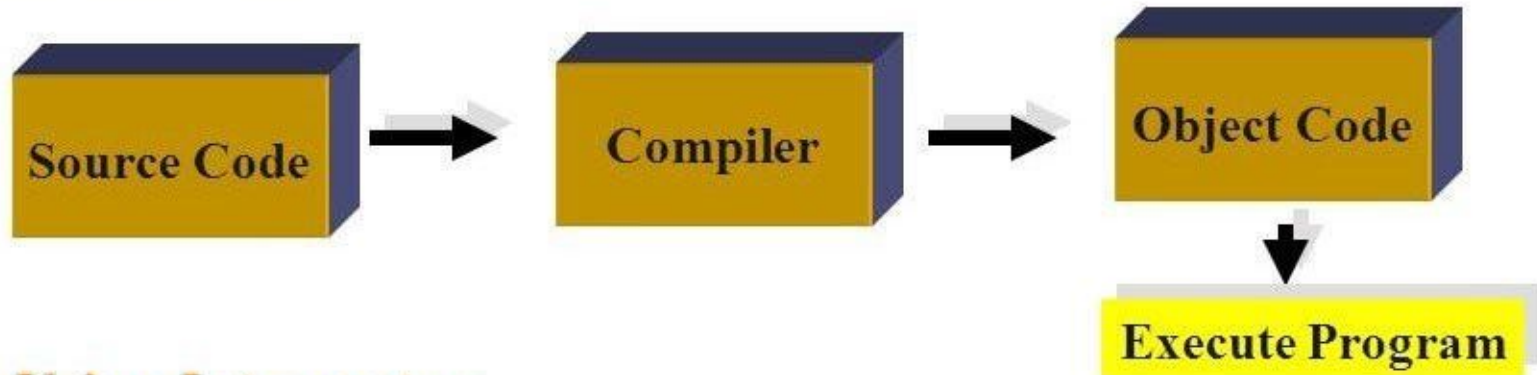
Examples of interpreted languages

- Python
- JavaScript
- Ruby

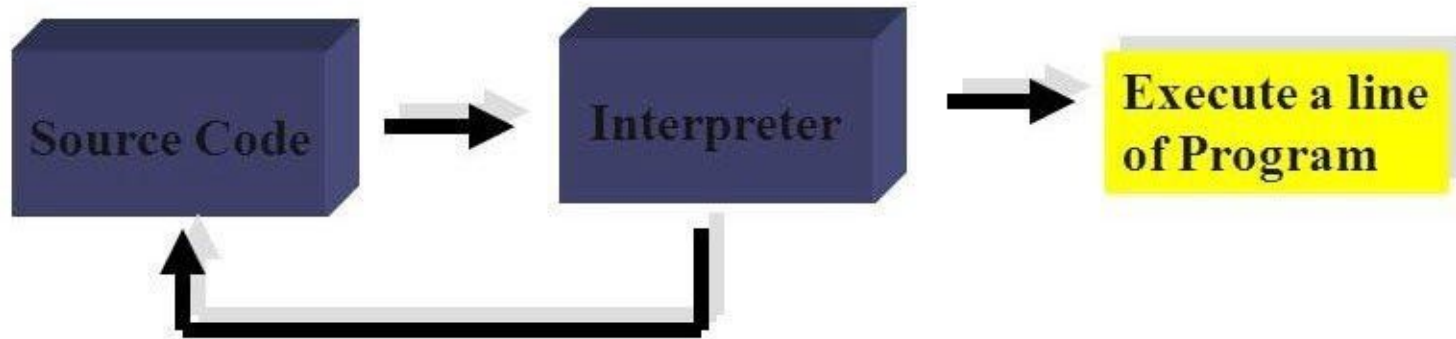
Example:

When you write a `print("Hello")` in python, the interpreter runs this line instantly.

- Using Compiler:



Using Interpreter:



Best of Luck for the Week 5 Tasks

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